

OIL INDIA LIMITED (OIL) - DULIAJAN HEADQUARTERS

DOCUMENT: OIL_UpperAssam_Barail_Loss_Study_2025.pdf (Research Paper & Technical Study)

WELL ID: FIELD-STUDY | FORMATION: Barail Coal-Shale (Regional) | INTERVAL: 3,380 m â 3,470 m

DATE: 2025-12-10 | AUTHOR: Center of Excellence for Energy Studies (CoEES) & Oil India R&D, Du

SHA-256 INTEGRITY HASH: 40b45217a2ae6235e617cef53f8813fd50da6c870b8ffe0d9bbc01bcf5139131

EXTRACTED EVENT: Regional Cleat Fracture Loss & Wellbore Instability @ 3400 m

SEVERITY: HIGH | LOSS RATE: 14.5 bbl/hr

MITIGATION: Proactive Stress-Caging at 3,385 m using bimodal CaCO3 (50/150 micron) + Nut-Plug

SUMMARY: Geomechanical & Cleat-Permeability Study of Oligocene Barail Coal-Shale across 24 Wells in Greater Duliajan and Nahorkatiya Fields. VERBATIM OPERATIONAL LOG: EXECUTIVE FINDINGS (CoEES / OIL R&D): Statistical analysis of 24 directional wells in Block NHR-04 and NHR-05 reveals a 68% incidence of severe lost circulation (10â 22 bbl/hr) and subsequent coal-packoff stuck pipe within a narrow 50-meter depth window (3,400 m to 3,450 m MD) inside the middle Barail Coal-Shale member. Because Barail coal cleats possess aperture widths of 80â 220 microns, conventional bentonite mud fails to bridge until massive losses occur. Proactive wellbore strengthening ('Stress Caging') must be initiated 15 to 25 meters ABOVE the 3,400 m topâ specifically at 3,380â 3,390 mâ by staging a 25 bbl bimodal LCM pill (15 ppb Nut-Plug + 10 ppb Flake Mica + 8 ppb Sized CaCO3) and restricting ECD below 1.28 sg.

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